

Unnamed_Unit

Unit 2 - Brainstorming & Idea Generation

PDF Documents

PDF Document 1: 1738158432725-Chapter 12 - Unit 2 - Brainstorming & Idea Generation.pdf

This PDF document is included in the unit materials.



UNIT 2 BRAINSTORMING & IDEA GENERATION

Introduction

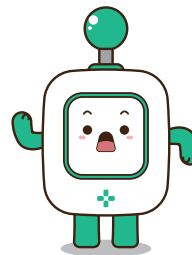
In Unit 2, your team will take the project idea you developed in Unit 1 and start turning it into reality. This involves planning how your smart device will work, selecting the MODI modules needed, and creating a detailed design. Your teamwork will be crucial as you move from ideas to action!



What You'll Do in This Unit

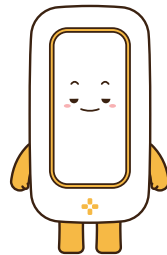
1. PROJECT PLANNING

- Work together to outline the steps needed to create your smart device, including a timeline and who will do what.



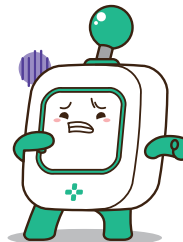
2. MODULE SELECTION

- Organize your algorithms logically. Break down complex tasks into smaller, manageable steps. Ensure that each part of the algorithm flows smoothly into the next, creating a clear and logical progression.



3. DESIGNING YOUR DEVICE

- Start sketching and discussing the physical design of your device, thinking about how it will look and work.



4. BUILDING A PROTOTYPE

- If time allows, begin assembling a simple prototype to test your ideas and make improvements.

Project Planning



Getting Ready to Build

Before you start building your smart device, it's important to plan out each step. Good planning helps your team stay organized and ensures that everyone knows what to do.



Outline the Steps:

- As a team, list all the tasks needed to complete your project.
- Think about the order of these tasks and what needs to happen first.
- Drawing Activity: Create a flowchart or diagram that shows the sequence of tasks. This visual plan will help your team stay on track.



Set a Timeline:

- Decide how long each task will take and set deadlines for completing them.
- Make sure your timeline is realistic and gives you enough time to do each step well.
- Drawing Activity: Use a calendar or timeline template to plot out the deadlines. This visual timeline will help you see how your project will progress over time.



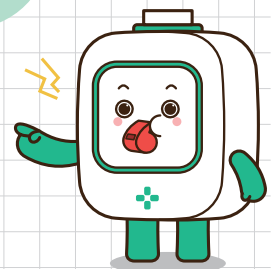
Assign Roles:

- Divide the tasks among your team members based on each person's strengths.
- Make sure everyone knows what they're responsible for.
- Drawing Activity: Create a role chart or team map that shows who is responsible for each task. This will help everyone stay aware of their roles and contributions.



Your Drawing Notes

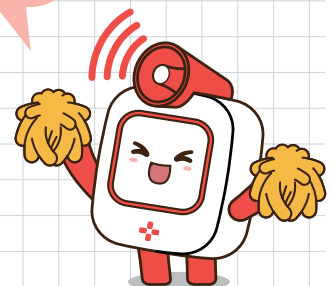
Think big! Don't be afraid to brainstorm even the craziest ideas. Sometimes the best solutions come from unexpected places.





Your Drawing Notes

Use diagrams and sketches to visualize your ideas and make your planning clearer.





Final Thoughts

Reflecting on Your Progress

Congratulations! Your team has made significant progress by turning your project idea into a detailed plan and beginning the design process. You've worked together to outline the necessary steps, chosen the right MODI modules, and started designing your smart device. These steps have laid a strong foundation for the next phase of your project.

1. Project Planning:

- You created a clear plan with a timeline, task assignments, and visual aids like flowcharts and timelines to guide your work.

2. Module Selection:

- Your team carefully selected the MODI modules needed for your smart device, ensuring that each one serves a specific function in your design.

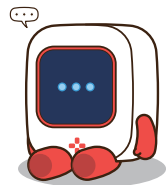
3. Designing Your Device:

- You sketched and refined your device's design, considering both function and user interaction.
- Your team collaborated to create a final design that everyone is excited to build.

4. Building a Prototype:

- If you started building a prototype, you learned valuable lessons from testing and adjusting your design.

Final Thoughts



In the next unit, your team will dive deeper into building your smart device. You'll refine your design, assemble the final product, and begin programming it to work as intended. The work you've done in this unit will guide you through the challenges ahead, and your strong teamwork will continue to be your greatest asset.